



AI-DRIVEN

CYBER SECURITY

*AI AGENTS ARE POWERFUL, BUT WHAT IF EVERY AGENT COULD
SECURELY CONNECT TO YOUR TOOLS, DATABASES, APIS, AND
SECURITY SYSTEMS THROUGH ONE STANDARD PROTOCOL?*

WHAT IS MCP?

- MCP = MODEL CONTEXT PROTOCOL
- AN OPEN STANDARD FOR CONNECTING AI TO EXTERNAL TOOLS AND DATA
- GIVES AI AGENTS A STANDARD WAY TO USE TOOLS
- WORKS WITH DIFFERENT AI MODELS AND APPLICATIONS
- THINK OF MCP AS A UNIVERSAL CONNECTOR FOR AI

MCP CLIENT & MCP SERVER

- CONNECTS THE AI APPLICATION TO MCP SERVERS
- SENDS REQUESTS TO MCP SERVERS
- RECEIVES TOOLS AND RESULTS
- PROVIDES TOOLS, RESOURCES, OR PROMPTS
- RUNS THE ACTUAL TOOL/FUNCTION
- CONTROLS WHAT THE AI CAN ACCESS
- CAN CONNECT TO APIS, DATABASES, FILES, AND SECURITY TOOLS

CONFIGURING MCP CLIENTS & SERVERS

- CLIENT CONNECTS TO AN MCP SERVER
- SERVER EXPOSES SPECIFIC CAPABILITIES
- TOOLS HAVE NAMES, DESCRIPTIONS, AND INPUTS
- CLIENT DISCOVERS AVAILABLE TOOLS
- AGENT CAN THEN REQUEST THOSE TOOLS

SECURE TOOL ACCESS

- NEVER GIVE AN AI UNLIMITED ACCESS
- EXPOSE ONLY REQUIRED TOOLS
- VALIDATE TOOL INPUTS
- AUTHENTICATE SENSITIVE SERVICES
- LOG IMPORTANT TOOL ACTIONS
- SEPARATE READ AND WRITE OPERATIONS

PERMISSION MODELS

- READ-ONLY ACCESS IS SAFER
- SENSITIVE ACTIONS NEED STRONGER PERMISSIONS
- DESTRUCTIVE ACTIONS SHOULD REQUIRE APPROVAL
- PERMISSIONS SHOULD BE BASED ON THE AGENT'S TASK
- READ . ANALYZE . WRITE . EXECUTE . ADMIN

MULTI-STEP REASONING

- COMPLEX TASKS NEED MULTIPLE DECISIONS
- AGENT PROCESSES ONE STEP AT A TIME
- EACH RESULT CAN INFLUENCE THE NEXT STEP
- TOOLS CAN BE USED BETWEEN STEPS

MANAGING MCP RISKS

- MCP CREATES POWERFUL CONNECTIONS, SO SECURITY MATTERS.
- MALICIOUS OR UNTRUSTED MCP SERVERS
- EXCESSIVE TOOL PERMISSIONS
- PROMPT INJECTION THROUGH TOOL DATA
- SENSITIVE DATA EXPOSURE
- MALICIOUS TOOL ARGUMENTS
- COMPROMISED CREDENTIALS
- UNSAFE TOOL EXECUTION